

EXPERIMENT – 9

Title

To Make Views of a Table

Aim / Objective

To understand the concept of **Views** in Oracle SQL and create, retrieve, update, replace, and drop views on database tables for simplifying data access and enhancing security.

Theory

A **View** is a virtual table that is created using the result of an SQL query. Unlike a physical table, a view does not store data permanently; instead, it displays data retrieved from one or more underlying tables whenever it is accessed.

Views help simplify complex queries, improve security by restricting access to selected columns or rows, and provide logical abstraction over database tables.

Advantages of Views

- Simplifies complex SQL queries.
 - Provides data security by restricting access.
 - Hides unnecessary columns.
 - Improves data abstraction.
 - Provides customized views for different users.
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Types of Views

1. Simple View

Created from a single table.

2. Complex View

Created using multiple tables, joins, aggregate functions, or group operations.

Learning Objectives

After completing this experiment, students will be able to:

- Understand the concept of views.
 - Create simple and complex views.
 - Retrieve data using views.
 - Update data through views.
 - Replace and delete views.
 - Apply views for database security and report generation.
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Software Requirements

- Oracle Database
 - Oracle SQL Developer / SQL*Plus
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Algorithm / Procedure

1. Open Oracle SQL Developer.
 2. Connect to the Oracle database.
 3. Create the Employee table.
 4. Insert sample records.
 5. Create different types of views.
 6. Retrieve data using views.
 7. Modify and delete views.
 8. Observe the output.
 9. Stop.
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Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee
(
EmpID NUMBER PRIMARY KEY,
EmpName VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER(10,2)
);
```

Step 2 : Insert Sample Records

```
INSERT INTO Employee VALUES(101,'Rahul','IT',35000);  
INSERT INTO Employee VALUES(102,'Priya','HR',42000);  
INSERT INTO Employee VALUES(103,'Aman','Finance',39000);  
INSERT INTO Employee VALUES(104,'Sneha','Marketing',45000);  
COMMIT;
```

Program 1 : Create a Simple View

```
CREATE VIEW Employee_View  
  
AS  
  
SELECT EmpID,  
EmpName,  
Department  
  
FROM Employee;
```

Display View

```
SELECT * FROM Employee_View;
```

Expected Output

EmpID	EmpName	Department
-------	---------	------------

101	Rahul	IT
102	Priya	HR
103	Aman	Finance
104	Sneha	Marketing

Program 2 : Create a View with Condition

```
CREATE VIEW IT_Employee
```

```
AS  
  
SELECT *  
  
FROM Employee  
  
WHERE Department='IT';
```

Display View

```
SELECT * FROM IT_Employee;
```

Expected Output

EmpID	EmpName	Department	Salary
101	Rahul	IT	35000

Program 3 : Create a Complex View

```
CREATE VIEW Salary_Report  
  
AS  
  
SELECT  
  
Department,  
  
AVG(Salary) AS Average_Salary  
  
FROM Employee  
  
GROUP BY Department;
```

Display View

```
SELECT * FROM Salary_Report;
```

Expected Output

Department Average Salary	
IT	35000

Department Average Salary

HR 42000

Finance 39000

Marketing 45000

Program 4 : Update Data Using View

UPDATE Employee_View

SET Department='Accounts'

WHERE EmpID=102;

COMMIT;

Verify

SELECT * FROM Employee;

Expected Output

EmpID	EmpName	Department
-------	---------	------------

101	Rahul	IT
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102	Priya	Accounts
-----	-------	----------

103	Aman	Finance
-----	------	---------

104	Sneha	Marketing
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Program 5 : Replace View

CREATE OR REPLACE VIEW Employee_View

AS

SELECT

EmpID,

EmpName,

Salary

FROM Employee;

Display View

SELECT * FROM Employee_View;

Expected Output

EmpID	EmpName	Salary
-------	---------	--------

101	Rahul	35000
-----	-------	-------

102	Priya	42000
-----	-------	-------

103	Aman	39000
-----	------	-------

104	Sneha	45000
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Program 6 : Drop View

DROP VIEW Employee_View;

Verify

SELECT * FROM Employee_View;

Expected Output

ORA-00942: Table or View does not exist.

Flowchart

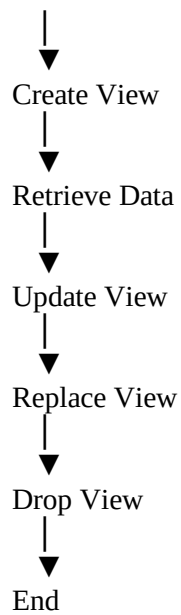
Start



Create Table



Insert Records



Explanation

CREATE VIEW

Creates a virtual table based on an SQL query.

Simple View

Created using a single table.

Complex View

Created using joins, aggregate functions, or multiple tables.

CREATE OR REPLACE VIEW

Modifies an existing view without dropping it.

DROP VIEW

Removes the view from the database.

Updating Through a View

If the view is based on a single table and does not contain aggregate functions, DISTINCT, or

GROUP BY, data can usually be updated through the view.

Observation

- Simple and complex views were created successfully.
 - Data was retrieved using views.
 - Records were updated through the simple view.
 - The view was modified using the CREATE OR REPLACE statement.
 - The view was deleted successfully.
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Result

Views were successfully created, modified, queried, and deleted. The experiment demonstrated how views simplify data access, provide security, and support logical data abstraction in Oracle databases.

Applications

- Employee Management Systems
 - Banking Applications
 - Hospital Management Systems
 - Payroll Systems
 - Student Information Systems
 - Inventory Management Systems
 - Report Generation
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Advantages

- Simplifies complex SQL queries.
 - Improves data security.
 - Restricts access to sensitive information.
 - Enhances code reusability.
 - Provides logical independence.
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Limitations of Views

- Complex views cannot always be updated.
 - Performance may decrease with nested views.
 - Views depend on the underlying tables.
 - Dropping the base table makes the view invalid.
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Viva Questions

1. What is a View?
 2. What is the difference between a table and a view?
 3. What is a Simple View?
 4. What is a Complex View?
 5. Can data be updated through a view?
 6. What is the purpose of CREATE OR REPLACE VIEW?
 7. What is the difference between a View and a Materialized View?
 8. What happens if the base table is dropped?
 9. Why are views used in database security?
 10. What are the advantages of using views?
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Lab Assignment

1. Create a view showing employee names and salaries only.
2. Create a view displaying employees with salaries greater than ₹40,000.
3. Create a department-wise salary report using a view.
4. Update employee records through a simple view.
5. Replace an existing view with additional columns.
6. Drop an existing view.
7. Compare simple and complex views.
8. Explain the advantages of views in database security.
9. Create a view using multiple tables.

10. Explain the difference between a View and a Materialized View.

Precautions

- Ensure the base table exists before creating a view.
 - Use meaningful names for views.
 - Avoid updating complex views containing aggregate functions or GROUP BY.
 - Verify the output after creating or modifying a view.
 - Remember that dropping the base table invalidates dependent views.
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Learning Outcome

After completing this experiment, students will be able to:

- Create and manage simple and complex views in Oracle.
- Retrieve and update data using views where applicable.
- Modify and remove views using SQL commands.
- Understand the advantages and limitations of views.
- Apply views to simplify queries, improve security, and provide logical data abstraction in real-world database applications.